3 Reconfiguring the Remainder: Agonistic Encounters with Social Robots

Whether they are used for personal care or welding cars, robots epitomize complex engineered systems. They weave together software and hardware; interface, interaction, and industrial design; and mechanical engineering, electrical engineering, and computer science. They employ advanced technologies, appear as and work in diverse forms and contexts, and play a role in popular and scientific histories and trajectories. Robots are, therefore, another notable category of computational objects through which to explore what it means to do design with computation.

However, the technical answer to the question "What is a robot?" is a matter of considerable debate. In computer science and engineering, the answer to the question has disciplinary significance and marks borders between conflicting approaches to operationalizing nontrivial subjects such as perception, affect, and cognition. Much of the debate in defining robots is traced to differing notions of intelligence—which is considered to be a fundamental property of a robot. Scientists from a classical artificial intelligence (AI) perspective generally argue that a robot requires the capacity for symbol manipulation and the possession of a symbolic model or representation of the world.[1] In contrast, those scientists who endorse what is sometimes referred to as "nouvelle AI" generally counter that intelligence is not synonymous with symbol manipulation and that a robot does not need a model of the world, as "the world is its own best model. . . . The trick is to sense it appropriately and often enough" (Brooks 1990, 6). At stake in this debate are the questions of what constitutes or what counts as intelligence and how to construct a computational artifact or system that can be claimed to possess some form of intelligence.

But intelligence alone does not answer the question "What is a robot?" In addition to the attribute of intelligence, physicality is commonly considered to be a fundamental property of a robot. Consider that virtual on-screen characters are referred to as agents, not robots, even though they may display qualities of intelligence, and yet a physical artifact, such as a toaster or a vacuum cleaner, that displays even rudimentary qualities of intelligence is often referred to as a robot.

In a discussion of design, these qualities can be drawn together in a simple and direct manner to answer the question "What is a robot?" When computational intelligence and the physicality of an artifact are bound together, we have a thing that can be called a robot. This binding of computational intelligence and physicality is significant because it constitutes a kind of embodiment, which structures the possible relations between people and robots. This embodiment makes robots distinctive as a category of computational objects. But the embodiment of robots is not naturally occurring; it is designed. A robot's embodiment is a consequence of how "intelligence" and "the artifact" are purposefully brought together.

There are many types of robots, including industrial robots, military robots, medical robots, service robots, and social robots. Of these, social robots present a fascinating yet awkward set of issues for contemporary design and have novel political concerns attached to them. Social robots are distinguished from other classes of robots by their modes of interaction and their purposes. They are designed to engage in communicative exchanges with people; to serve human needs beyond those of labor or common notions of work; and to operate with individuals and small groups of people in homes, in health care facilities, or out in public. Most social robots exist as not-quite products; as artifacts in a liminal state between academic and industrial research labs and the consumer market. But at consumer-good expositions where corporations exhibit their near future wares, social robots are increasingly present. Designers are exploring and experimenting with new forms and modes of interaction for social robots. These explorations and experiments evoke political issues: the ways in which we design the character of our relationships with social robots reflect and reinforce beliefs about what it means to be social and set trajectories for how we might live together with computational artifacts in an increasingly intimate manner.

As an example, consider PARO, the baby seal therapy robot—one of the few social robots that exist as a commercially available product.[2] PARO is encased in antibacterial fur and designed for physical interaction with humans. It responds to touch, sound, and the presence of others through changes in its body position and by making sounds similar to the animal it imitates. It perceives aspects of the environment and adjusts its behavior accordingly—for example, sleeping when the lights are off. It is also able to learn the preferences of its users over time and move and communicate in ways that will be most pleasing and beneficial to them. Stroking PARO functions as positive behavior reinforcement, and it will register the last actions it did before the stroking occurred and later repeat them. Likewise, striking PARO functions as negative behavior reinforcement, and it will register the last actions it did before the striking occurred and later will not repeat them. Under the fur of the robot are sensors that monitor light, sound, touch, and the position and movement of objects around the robot. Data from the sensors are registered and processed, and instructions are sent to a suite of motors to move accordingly and to play sounds from a speaker embedded below the fur surface of the robot. Sensors detect environmental factors as they change nearly continuously (people move, shadows fall, sounds increase and decrease in volume), and processing and actuation are repeated over and over, resulting in an exhibition of animation and interactivity.

The name PARO is derived from the phrase *personal robot* and immediately associates PARO with the category of social robots. In fact, PARO is described by its designers as a "mental commitment robot" and is designed to elicit emotional response and attachment from users.[3] For example, one scenario of use for PARO is as a surrogate in animal therapy, functioning in a manner analogous to a service or companion animal by providing cognitive and emotional support. The underlying idea is that users will interact with PARO and develop a relationship with the robot similar to the kinds of relationships developed between people and animals.[4]

PARO is not just a trivial gadget or an obscure technological showpiece. Substantial funding and intellectual effort have been put toward the development of this robot. Interactions with it have been studied from multiple methodological perspectives to ascertain its psychological, physical, and social effects.[5] The research teams at the National Institute of Advanced Industrial Science and Technology, who designed PARO, and others have produced scholarly publications related to the robot.[6] It is also available for sale from PARO Robots, Inc., a corporation developed to move PARO from the lab into the market. By most common metrics, PARO is as real and legitimate as any other new technology product.

The design of PARO provides one set of answers to the question "What might be the character of our future relationships with robots?" The embodiment of PARO, carefully crafted through the design of form, materials, behavior, and expression, structures a particular set of possible relations between people and the robot. In its marketing literature, PARO Robotics, Inc. asserts precisely such a connection between the design of the robot and the kinds of relationships it is intended to induce (PARO Robotics U.S., Inc. 2008):

> Covered in pure white synthetic fur, the built-in intelligence provides psychological, physiological, and social effects through physical interaction with human beings. PARO not only imitates animal behavior, it also responds to light, sound, temperature, touch and posture, and over time develops its own character. As a result, it becomes a "living" cherished pet that provides relaxation, entertainment, and companionship to the owner.

Through such descriptions and the designs that accompany them, the association of terms such as *social*, *living*, and *companionship* to PARO concomitantly defines the robot and redefines these terms in regard to the robot. That is, by labeling the robot as social, we have certain expectations of it and of our interactions with it. At the same time, the label *social* also takes on new meaning in considering computational objects as entities that people are or might be social with.

Reflecting on PARO as a designed thing prompts questions and issues concerning how to use robots and what role should be played by design in shaping our experiences with these computational objects. But PARO is not an example of adversarial design. PARO is emblematic of what would be designed against and of what adversarial design works to question, challenge, or resist. Before proceeding to specific tactics of designing agonistic encounters with social robots, it is worthwhile to examine a bit more the political issues of social robot design.

The Political Issues of Social Robot Design

Science studies scholar Lucy Suchman (2006, 239) draws attention to a set of inherently political concerns with social robots when she states, "For me, however, the fear is less that robotic visions will be realized (though real money will be diverted from other investments), than that the discourses and imaginaries that inspire them will retrench received conceptions both of humanness and of desirable robot potentialities, rather than challenge and hold open the space of possibilities." This notion of retrenching is vivid and should be taken literally to express the ways that lines are drawn and positions defended about what counts as proper and preferred relations between people and robots. One way these lines are drawn is through design—through the making of robots that materialize and enact particular conceptions "of humanness and of desirable robot potentialities." Suchman is not arguing against robots but rather calling attention to the need to examine assumptions within robot design and consider alternatives. If we want robots as companions, what kinds of companionship do we want to engage in with them? What models of human companionship or sociability are we drawing from and designing into these machines, and are these really the models we want to emulate, or should others be considered and designed for?

The question for design is not whether to engage in social encounters with robots but rather *how* to engage in social encounters with robots. Will the design of these encounters reinforce reductive and staid notions of what it means to be human? Will the design cover over anxieties brought about by such animated technology? Or will the design agonistically engage these concerns and perhaps even suggest new experiences with robots?

Consider the baby seal robot PARO again. Throughout video demonstrations, marketing materials, and research papers, it is presented as an innovative and feasible technological solution to the problem of providing therapy to those in need.[7] But using an animated intelligent machine for personal, mental comfort is not a casual, everyday scenario.[8] This animated intelligent machine imitates an animal that would otherwise rarely come into contact with people. One might expect that the strangeness of the situation would confuse people who were presented with the proposition of interacting with PARO. But the design of PARO mitigates such responses. The seal-likeness of the robot is itself a caricature, more like a child's stuffed animal or toy than an actual creature. The design of the robot (as something cute and docile, with soft fur, wiggling motions, and purring sounds) and the user's interaction with it (as a tactile affair in which users hold and stroke the robot as it sits in their laps) moderates the unusual scenario of seeking solace from a machine. Through its design, PARO, which materializes and enacts one idea of human-robot relations, is made to appear pleasing, advantageous, and relatively without issues.

The question of how people will relate to and interact with robots, however, is an issue. Surfacing this question and exploring the issues that underlie it can be agonistic endeavors in the sense of agonism as an activity of ongoing contest between ideas through which dominant perspectives and assumptions are revealed and critiqued (Mouffe 2000a, 2005b). The design of social robots can be interpreted as a political issue—and as an activity of design with political qualities—because through shaping encounters between people and robots, expectations and norms concerning those relations are established and reinforced. These expectations and norms have lasting effects. As Suchman (2006) notes, they influence research and product development trajectories, which are enforced by allocations of funding and acceptance in both academic and market settings. The design of social robots also shapes how we understand concepts like care, which may, in turn, affect how we develop other products and services. This issue is addressed at length by another science studies scholar, Sherry Turkle, who has worked extensively with the robot PARO and raises moral and ethical questions concerning social robots that reflect perspectives on what it means to be human and the nature of the human experience. In some cases, these questions have clear political qualities and implications, such as when Turkle asks, "Do plans to provide relational robots to children and the elders make us less likely to look for other solutions for their care?" (Turkle 2006, 2).

This is a different kind of political issue and expression from what has been discussed so far in this book. The political qualities of social robot design are not as immediately obvious as the political qualities of campaign finance or the price of oil. The political qualities of social robot design concern personal relations between ourselves and others and questions about how design shapes these relations. The implications of these issues and the consequences of design lie more in the future than the present, as social robots are still mostly a class of products in development. Addressing the political issues of social robot design is important because it demonstrates how design can be preemptive in its political provocations to engage issues further upstream in the research and development process. The question is, How can design do the work of agonism in the context of social robots?

Designing Agonistic Encounters with Social Robots

Agonistic encounters with social robots work to expose perspectives and assumptions in robot design and make veiled issues and excluded possibilities available for inquiry and critique in material form. Through such agonistic encounters, critical perspectives on human-robot relations are put forth. They thus work to keep open the space of possibilities for design and for our future interactions with robots, allowing for a pluralism of design engagements. I characterize the endeavor of designing such agonistic encounters as *reconfiguring the remainder*, bringing together Lucy Suchman's notion of reconfiguration with political theorist Bonnie Honig's notion of "the remainder."

In *Human-Machine Reconfigurations: Plans and Situated Actions*, Suchman (2006) proposes reconfiguration as a tactic for rethinking the design of computational systems and our interactions with them. Suchman's notion of reconfiguration is developed from feminist approaches to science and technology studies, particularly those of Donna Haraway, positing that "technologies are forms of materialized figuration; that is, they bring together assemblages of stuff and meaning into more or less stable arrangements" (Suchman 2006, 226). Following from this idea, according to Suchman (2006, 226), "One form of intervention into current practices of technology development, then, is through a critical consideration of how humans and machines are configured in those practices and how they might be figured—and *configured*—differently." This notion of configuration and reconfiguration refers both to the technological organization of the system and to the relationship between people and technical systems.

Increasingly, the design of computational systems and products is as an activity of innovation through configuration. Although some aspects of a computational system may be newly invented, most often design produces a custom arrangement of parts, capacities, affordances, and concept to achieve some desired result. For example, the design of a consumer product such as the Apple iPod can be understood as the configuration of various sensors for gestural interaction, a touch-screen display, hard-drive storage and access, and the concept of mobile personal entertainment. Likewise, the design of PARO can be taken as the configuration of various haptic, vision, and auditory sensors, actuators, fur and the baby seal form, and the concept of therapy. Each particular configuration structures, by design, specific kinds of relations between users and the artifact.

Drawing from Suchman, I extend her term *reconfiguration* to describe an agonistic alternative to the design of computational objects. With agonistic reconfiguration, the object is still designed by a custom arrangement of parts, capacities, affordances, and concept. But it is done in a provocative manner that purposefully deviates from familiar configurations. The agonistic activity of reconfiguration is the combining of components and concepts together in unexpected, exaggerated, or otherwise purposefully atypical ways, which produce disjunctions between expectations, the material artifact or system, and the experience of it. Such provocative reconfigurations are not accidental or arbitrary. Rather, the activity of reconfiguration leverages an understanding of the standards of configuration, both technically and socially. It works by manipulating those standards and addressing what is left out of common configurations, which can be referred to as "the remainder."

Political theorist Bonnie Honig uses the term *remainder* to describe what is expelled in politics. This term refers to the people, practices, and discourses that are overlooked or written out of institutions, policies, legislation, and theories in the attempt produce a consensus that lacks conflict or disruptive differences. But under every condition and from every political position, something is excluded. As Honig (1993, 5) states, “All sets of arrangements are invariably troubled by remainders.” One agonistic endeavor is to identify what has been excluded and ask, Why?, and, How would its inclusion reconstitute a given condition or thing?

This excluded remainder can also refer to the veiled issues and excluded qualities that are overlooked, written out, or otherwise expelled from designed things. *Reconfiguring the remainder is an agonistic tactic of including what is commonly excluded, giving it privilege, and making it the dominant character of the designed thing.* In the case of social robots, rather than using design to hasten the acceptance of a social robot or to settle the question of how people and robots will interact with one another, reconfiguring the reminder gives critical pause to this process of social integration.

Embodiment

A political analysis and critique of social robots could pursue many factors, for instance, form, function, interaction, and experience are the commonplaces of design and design criticism. These factors, however, are surface expressions of the principal quality that makes social robots distinctive as a category of computational objects. That is, form, function, interaction, and experience are outcomes or effects of the design of a robot’s embodiment. To understand how embodiment is treated in the design of agonistic encounters with robots—how it evokes political issues and can be interpreted politically—requires first clarifying how embodiment can be understood as a designed quality of an object.

Unlike procedurality, a quality that is easily attributable to the medium of computation, claiming embodiment as a defining quality of robots is difficult because it is often construed as a quality that is distinct to living entities. Theories of embodiment from phenomenology and embodied cognition have woven their way into robotics research through the fields of artificial intelligence and the associated questions about what is required for knowing and acting in the world.[9] Embodiment in robotics is similar to notions of embodiment in philosophy and cognitive science, where embodiment is concerned with the body of an entity and its capabilities in defining being in the world. But within robotics discourse, embodiment is not limited to living entities: it can just as easily be a quality of a nonliving entity.

The particular characteristics and effects of embodiment are a significant topic of research in the field of human-robot interaction. Roboticist Kerstin Dautenhahn is one the scientists at the forefront of this research, and her work is useful for framing an understanding of embodiment beyond living entities. As Dautenhahn and her coauthors state, in terms as equally applicable to a robot or human, embodiment can be characterized as "that which establishes a basis for structural coupling by creating the potential for mutual perturbation between system and environment" (Dautenhahn, Ogden, and Quick 2002, 400). In her research, Dautenhahn has been concerned in particular with embodiment from an experimental science perspective. She has developed a framework for embodiment that can be operationalized and quantified, such that degrees of embodiment might be empirically compared among different entities in different environments, with different capabilities and affordances. From this effort, she has developed a definition to describe the state of embodiment within any system: (Dautenhahn, Ogden, and Quick 2002, 400):

A system *S* is embodied in an environment *E* if perturbatory channels exist between the two. That is, *S* is embodied in *E* if for every time *t* at which both *S* and *E* exist, some subset of *E*'s possible states with respect to *S* have the capacity to perturb *S*'s state, and some subset of *S*'s possible states with respect to *E* have the capacity to perturb *E*'s state.

Even when the goal is not the quantitative measurement and comparison of embodiment, such a definition is useful for reframing common notions about embodiment. Such a definition enables moving beyond the notion that embodiment is a quality limited to living beings. As the variables in the definition denote, embodiment is particular-to-particular structures. Each configuration of a robot establishes a different set of possible couplings and mutual perturbations with the environment. Each environment or entity is likewise characterized by certain qualities and affordances that engage these configurations differently, resulting in a diversity of kinds of embodiment. Extending embodiment beyond living entities alone to include the artificial is significant because embodiment can therefore be treated as a quality that can be shaped and manipulated by design—by the choice and arrangement of particular aspects of the robot, including its hardware; software; capacities for sensing, processing, and actuation; form; and behavioral qualities.

But the form and structure of embodiment alone do not shape encounters with robots. Context must be considered with embodiment. The meaning of different kinds and experiences of embodiment will be construed based on the broader sociocultural contexts in which the robot is set. The home is not the battlefield; service animals are different from pets; seals are not familiar domestic creatures. So even with Dautenhahn's operational definition, embodiment is far from a reductive mechanistic quality. It is highly contingent. Different configurations or reconfigurations of embodiment produce different human-robot encounters in different contexts depending on expectations and desires in these situations. Recall the design of PARO's embodiment—soft fur, purring sounds, and wriggling motions in response to being stroked. Such a design may be appropriate for the contexts of the home, retirement center, or hospital but not to comfort wounded soldiers on the battlefield.

Since embodiment makes robots distinctive as a category of computational objects, it is a promising site of agonism. Designed embodiments and the subsequent forms, functions, and interactions that run counter to our expectations produce deviations from the norm in social robot design. These deviations result in experiences that challenge the familiar and proverbial in social robot design and expose topics of debate concerning our future relationships with robots. Examples of such designs and encounters often lurk just outside the established fields of interaction design and human-robot interaction. Suchman and others have drawn attention to such examples as sites of inquiry.[10] My intention is to build on those analyses through the frame of agonism to draw out design exemplars of social robots that evoke political issues and relations.

Engineering the Uncanny

Blendie is a kitchen mixer that the user speaks to and that speaks to the user (Dobson 2007a).[11] To interact with *Blendie*, users mimic the sounds made by a common kitchen mixer, and *Blendie* responds with a mechanical rendering of those sounds repeated back to the user, produced by varying the speed of its motor. As depicted in the project documentation,[12] a person approaches *Blendie* and begins to make all variety of grunting and whining machine-like sounds of high and low pitch. This instigates a response from *Blendie* of either fast or slow rotation of the motor, which produces a corresponding whirring sound. Over time, the back and forth of sounds uttered and imitated between *Blendie* and the user begins to suggest a dialog of sorts, albeit a strange one.

In an encounter between *Blendie* and a user, the user begins to transform herself to be more machine-like to elicit a response from the mixer.

Figure 3.1
Kelly Dobson, *Blendie* (2007a)

Adjusting one's modes of behavior to elicit responses from others is common and is fundamental for effective communication. Moreover, people often adjust their behavior to make use of or otherwise interact with machines—for example, slowing one's pace as approaching automatic doors to allow time for the system to note one's presence and respond by opening the doors. What is uncommon and arresting with *Blendie* is the requirement of dramatically adjusting human communicative modes with machines to, in effect, perform machine behavior.

This encounter between a person and a robotic kitchen mixer was designed by Kelly Dobson. *Blendie* is part of a larger body of work by Dobson titled *Machine Therapy*, which "tweaks technological artifacts in order to explore their sensitive and emotional side" (Dobson 2007b). Like PARO the baby seal robot, *Blendie* is designed for therapy. But the purposes and modes of therapy between these two robots are remarkably different. Whereas PARO's therapeutic purpose is alleviating human loneliness and remedying a perceived lack of affective exchange, *Blendie's* therapeutic purpose is the reflective exploration of human relationships with machines. With *Blendie,* the subject of therapy with the device is our relationship with devices. In addition, the mode of therapy advanced by Dobson's design is in stark contrast with that of PARO. Whereas PARO is designed to elicit and support a placid, soothing experience of therapy grounded in amity as healing, *Blendie* is designed to support a model of therapy grounded in unease and confrontation as the cure—the model of psychoanalysis (Dobson 2007a).

The *Machine Therapy* project is agonistic in that it provides alternate modes of interacting with robots that bring to the fore, rather than mitigate, the tension and anxiety that frequently characterize our relationships with technology. Throughout entertainment media, robots are often used to signal and explore these tensions and anxieties in amplified form. In many film representations of robots—Maria in *Metropolis* (1927), Deckard in *Blade Runner* (1982), the *Terminator* series (1984, 1991, 2003, 2009), the artificial boy David in *Artificial Intelligence: AI* (2001)—the robot is a figure at odds with its identity and our relationship to it. In *Metropolis*, it is seductively attractive yet inhumane and despotic; in *Blade Runner*, it is self-loathing; in the *Terminator* series, it is a brutal assassin transformed into a brutal savior; and in *Artificial Intelligence: AI*, it is a discarded anthropomorphic appliance. Exploring these tensions and anxieties through an object such as *Blendie* continues the tradition of the robot as a kind of reflective other but with an opportunity for interaction unavailable through other forms of media. And although this tradition of the robot as a reflective other exists within the cultural representations of robots (in films, television, and fiction), it is conspicuously uncommon in actual social robot design.

In giving voice to the tension between people and technology, *Blendie* affectively manipulates that anxiety and plays on the uncanny, which is a particularly compelling trope for exploring human-robot relationships because it operates by troubling existing categories of form, function, purpose, and being. In his essay "Das 'Unheimliche,'" Sigmund Freud characterized the uncanny as an experience in which the familiar suddenly becomes strange, resulting in a sense of psychological fear (Freud 2003).[13] Freud explores several instances of the uncanny and the common themes that run through them. One theme is animism and anthropomorphism, or the attribution of lifelike, human qualities to inanimate objects. This theme is still explored today and is found in many films that cross the boundary between science fiction and thriller and present visions of computational systems gone awry. In *The Shaft* (2001), an intelligent elevator is out for revenge, and in *One Missed Call* (2008),[14] mobile phones and data networks are possessed by a malevolent entity. More generally than elevators with a vengeance or malevolent mobile phones, the uncanny can be taken to be the wearing away of the distinction between the real and the imagined, "when we are faced with the reality of something that we have until now considered as imaginary" (Freud 2003, 150).

The uncanny has also found its way into robotics research, through the notion of the the uncanny valley. In 1970, roboticist Mashiro Mori postulated that as a robot appears more human, our acceptance of it increases. This acceptance follows an upward curve until the robot's resemblance to a human being reaches what is called the uncanny valley—a conceptual space in which the resemblance between a robot and a human are almost identical, and the tension between this difference and sameness is disturbing (Mori 1970). Most often, the uncanny valley refers to the visual appearance of the robot, but it is not limited to the visual realm alone. Mori accounted for a more nuanced notion that extends visual appearance to include aspects of presence, behavior, and interaction—a more robust notion of the uncanny that spans various kinds of embodiment. Although often the visual appearance seduces us into thinking that an object is real or alive, usually other forms of embodiment produce the experience of the uncanny. Consider an example of the uncanny valley used by Mori—shaking hands with a corpse. We expect the hand to feel warm and supple, and yet it is cold and rigid, disturbingly contrary to our notions and personal experiences of bodies. Even though the uncanny valley has just begun to be systematically examined (as much as it can be), the idea has been perpetuated in the robotics community over the past several decades.[15] Generally, it is deemed a place to be avoided in the design of robots, particularly robots intended for interaction with humans, because it is seen as a hindrance to the acceptance of robots.

In the context of social robot design, the uncanny is a theme and sensation that reflects the remainder. What is veiled or excluded by design in social robots is the apprehension, confusion, and anxiety often experienced by people who encounter objects that feign to be something other than they are and that invite interaction with them in a personal, even intimate, manner. From an agonistic perspective, however, the reasons that most researchers and designers attempt to avoid the uncanny can be recast as reasons to induce it. Uncanny encounters with robots produce troubling engagements between intelligent artifacts and people. They prompt reflection on the nature and substance of the relations between people and robots.

What is experienced or witnessed with *Blendie* is a reconfiguration of robot therapy and human-robot relations that leverages the uncanny to produce an agonistic encounter. This encounter is agonistic in that the very anxieties and tensions between intelligent artifacts and people that are usually smoothed over by design are here, by design, made into the basis of the interaction. Interacting with *Blendie* occurs not through placid stroking, but rather, through an agitated exchange of growling at it, and its growling back.

This reconfiguration is materially and experientially enacted through the design of the robot's embodiment. For *Blendie*, Dobson developed audio sensors and software that were fitted within the housing of a commercial kitchen mixer. The audio sensors monitor and register sound (the noises made by humans as they grunt and whine at the machine), analyze the sound for frequency and pitch, and then translate those qualities of human sound into numeric values that can be used to vary the speed of the mixer's motor, which acts as *Blendie*'s projected voice. This may seem like a simple form of coupling, but it is not simplistic. It demonstrates a sophisticated understanding of how to sculpt embodiment by design through the medium of computation (figure 3.2). As Dobson (2007a, 80) explains:

> Blendie works by taking in the sound of a person interacting with it through a microphone and processing that sound on a computer running custom software written in C++. The program computes an STFT (short time Fourier transform) to detect the dominant pitch, and an FFT (fast Fourier transform) of this STFT to look for time-domain frequency modulation. If it detects modulation in a range that has been predetermined as a close human approximation to the rough guttural sound of the blender's motor, Blendie then is given the correct amount of power to allow it to spin at a speed that will produce the same dominant pitch of the person's voice. The power is adjusted using PWM (pulse width modulation) of the AC (alternating current) line supplying power to Blendie. The proper PWM for a given pitch is returned from a large lookup table in the software custom made for the blender. The software can tell a human voice from a blender sound, and thereby can keep Blendie from forever feeding back on itself, because a human imitation of a blender is very different from the sound of the blender itself.

Dobson's design evokes the uncanny by intentionally reconfiguring the standard mode of embodiment from human language to machine sounds. This inverts the common relationship of person to machine, in which the person is (at least theoretically) given prominence. Thus, the design of *Blendie*'s embodiment is a machine-centric embodiment. The basis for the coupling is in machine terms rather than human terms.

This shifting of the basis of embodiment draws attention to a we/they relationship. Within theories of agonism, the notion of a we and a they—of an us and a them—is central to establishing difference and an adversarial stance (Mouffe 2000a, 2005b). Through these categories of difference, distinctions between beliefs, values, and practices are organized and

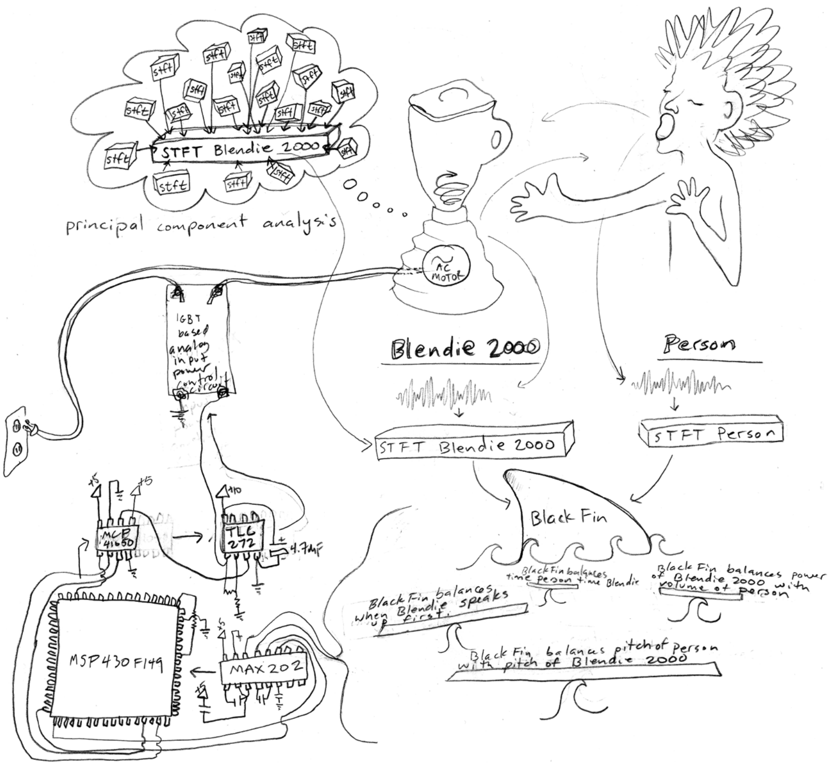

Figure 3.2
Sketch depicting the design of *Blendie*'s embodiment, Kelly Dobson (2007a)

expressed. The performance or acting out of the relationship between these categories, which is often one of tension, is the conflict that defines agonism and represents the political condition. However, the categories regarding social robot design are notably different. Rather than familiar distinctions of we and they, such as the ideological left and right, liberal and conservative, or pro and anti any given subject, here the we/they is, at least initially, a distinction between humans and machines. The tension that is identified and brought to the fore concerns how we conceptualize what is human and what is machine and how these conceptualizations inform and interact with each other.

But beyond just establishing those boundaries of and interactions between the we of humans and the they of robots, the agonistic endeavor in adversarial design is to investigate and question them as categories with political significance that are open to critique and reinterpretation. The challenge and opportunity of the uncanny as witnessed in *Blendie* goes beyond questions of whether a thing is or is not human, whether it is or is not alive. In fact, uncanny encounters transform the nature of we/they relations beyond initial simplistic categories of human and machine. Rather than rashly either valorizing those categories or making claims of dismantling them, one should instead consider the qualities that underlie those categories and the permeability between those categories. The political issue of concern for design with social robots is not registering or denying humanness. Rather, the political issue of concern for design is how people and robots are going to comingle.

An Uncanny Affective Companion

As computational systems have increased in sophistication and expanded in use, the fields of interaction design, human-computer interaction, and human-robot interaction have gained ground as important endeavors. These fields focus on the study and shaping of interactions with computational systems, often to understand patterns of use and meaning making in order to design systems that are useful, usable, and desirable. In mainstream human-robot interaction research and design, robots such as *Blendie* can be interpreted as provocations that question the base assumptions of such efforts. Recalling Suchman's (2006) concern about the "retrenching" of desires and imaginaries, such interventions are important because they challenge assumptions about modes of interaction with robots and thereby keep open the space of design possibilities, providing alternative themes for design.

The goal of productivity in the design and use of computational systems is an example of how themes in interaction design, human-computer interaction, and human-robot interaction develop, are materialized in systems, and are challenged and evolve. Much of the practice in these fields is geared toward improving the capacity of systems to enable people and industry to "get work done." Early in the history of the field of human-computer interaction, from the 1980s to the mid-1990s, the emphasis on productivity led to equating interaction with usability and usefulness with convenience and expediency. Productivity reigned as the predominant purpose of interaction design and human-computer interaction, against which research and practice were judged. Since the late 1990s, the singular importance of productivity has been steadily questioned through a stream of research projects and publications advancing alternative themes to drive the design and use of computational systems.[16] As a result of such efforts, today, *ludic*, *reflective*, and *pleasurable* are common qualifiers and themes against which many systems are judged.[17] This does not mean that the topic of productivity is resolved and that the contest over productivity is completed. From an agonistic perspective, the contest is never ended: one "needs to be always switching positions, because once any given position sediments, it produces remainders" (Honig 1993, 200). As soon as an issue appears to be settled, subsequent issues or positions emerge, and they need to be addressed. As the challenging of productivity has proceeded, it has developed connections with other themes. These provide other issues to be addressed and further sites of agonistic intervention: chief among them in regard to social robots is affect.

The theme of the robot as companion or partner (the terms are often used interchangeably in robotics discourse) is popular in robotics research and product development. Partner robots offer the potential of becoming a significant consumer market, and the robot as companion speaks to popular culture's connotations of robots, making them attractive for marketing and public relation purposes. PARO is one example of such a robot. Another example is the NEC PaPeRo. The name PaPeRo is derived from *partner-type personal robot*, and the robot was designed to be a research platform for personal robots for use in the home. Unlike many academic research robots, PaPeRo appears like a finished consumer product. It is made of plastic, brightly colored, and designed to look cute with gentle bulbous curves and large eyes. Over the course of its development, several roles have been identified for the PaPeRo. One of these roles is as a so-called childcare robot that functions as a companion to children and a partner to parents in the activity of parenting.[18]

With the exception of PARO, few such robots are readily available as products for either individual or institutional purchase. Most companion and partner robots are currently in the research and development phase. Regardless of whether a specific robot such as PaPeRo is used in the near future, its development and suggested use is evidence of a particular vision of the world in which robots and people work intimately together in their everyday lives. Along with this intimacy comes a series of expectations and standards of interaction. The design of these robots advances and materializes that vision for ongoing research and product development. Even though robots such as PaPeRo are not in mass commercial production, they still frame what can be expected of future robot products.

In the early history of computational systems design, there was little consideration of affect, which did not fit neatly with imperatives of productivity. This perspective has shifted in recent years as discourses of pleasure and play have gained prominence. In the design of companion robots, affect often takes on particular importance because it is assumed that for a robot to be an effective companion, it must take into account emotion. The phrase *affective computing* was coined by Rosalind Picard, a research scientist at the Massachusetts Institute of Technology, and has become something of a catchphrase for projects ranging from emotion models for machines to sensors that read the emotions of people. As Picard (2005, 3) describes her approach to affective computing:

> Affective computing includes implementing emotions, and therefore can aid the development and testing of new and old emotion theories. However, affective computing also includes many other things, such as giving a computer the ability to recognize and express emotions, developing its ability to respond intelligently to human emotions, and enabling it to regulate and utilize its emotions.

Affective computing is subject to philosophical debates that are similar to those concerning the nature of human and machine intelligence. These debates of definition influence the replication or simulation of certain qualities in artificial entities. Picard attempts to sidestep this issue by emphasizing the pragmatic ends of affect rather than the ontological status of emotion. In doing so, she reveals a common position in robotics that casts emotion as a necessity for achieving rationality and as a means for improving the productivity of the computational artifacts and systems. As Picard (2000, 280–281) states,

> The inability of today's computers to recognize, express, and have emotions severely limits their ability to act intelligently and interact naturally with us . . . because emotional computing tends to connote computers with an undesirable reduction in rationality, we prefer the term *affective computing* to denote computing that relates to, arises from, or deliberately influences emotions. *Affective* still means emotional, but may, perhaps usefully, be confused with effective.

Affect, in the context of robots and specifically social robots, is thus treated as a way of regulating a robot's behavior, as a quality of a machine's expression toward people, and as a way of regulating a person's encounter with a machine. The general idea is that affect, in the form of emotional models, improves the decision-making capabilities of robots and, in the form of expressive gestures, persuasively shapes desired interactions between people and robots.

From an agonistic perspective, this conceptualization of affect begs examination. What, in such conceptualizations of affect, is being left out?

What is the remainder of affect in the design of social robots? What alternative experiences of companionship and affect might be advanced through design?

Omo was created by Dobson in 2007 and, like *Blendie*, is a robot with a novel form of embodiment. *Omo* detects and responds to the breathing patterns of others and can deform its own structure to express breathing-like motions. *Omo* operates by monitoring the breathing patterns of those holding the robot through the use of pressure sensors. At times, *Omo* matches the breathing patterns of its companion; at other times, it offers a new pattern for its companion to match, guiding the user through a series of controlled breathing exercises.

The contrast between *Omo* and other social robots draws into relief some of the developing assumptions concerning the design of robots, particularly the design of robots as companions. Foremost is *Omo*'s form—egglike, glowing, and rubbery (figure 3.3). In contrast to the common design approach to such robots, it is not cute. Its appearance does not mimic a domesticated pet, it is not fuzzy, and it does not have baby face features, such as large eyes. In comparison to PARO, *Omo* appears alien. Even next to robots such as the NEC PaPeRo, which also has a squat rotund form, *Omo* is distinctive in its lack of anthropomorphic or zoomorphic features: it has no eyes and no mouth. In Dobson's (2008) own comments, *Omo* is designed to be more like an organ than an animal or a person.[19]

Figure 3.3
Kelly Dobson, *Omo* (2007b)

Furthermore, the use of breathing as the basis of embodiment—that is, as the source of sensed input and output—enhances the uncanny status of the robot. When robots are given lifelike qualities, it is usually via the common notion of form as shape and volume or the presence of specific anthropomorphic or zoomorphic features. For example, PARO appears lifelike because its appearance imitates an animal, and one instinctively anthropomorphizes PaPeRo due to the outline and position of the eyes in its head. To make the foundation of the robot's embodiment associated with the activity of breathing—an activity that is distinctive to living entities and only to living entities—is an uncommon design decision, which results in an uncanny experience of the robot. *Omo's* uncanny-ness, however, does not create distance from the computational object. It draws people closer in what is potentially a much more powerful affective relationship as it calls forth a psychologically complex form of communication and exchange through the experience of shared breathing. As Dobson (2008) describes the interaction with *Omo*, "as you are holding it, you will slowly change with it, much like as you hold another person you start breathing together." Even compared with the tactile interaction of PARO, the breathing together with *Omo* appears strikingly intimate. Furthermore, whereas with PARO the design is an attempt to mitigate any strangeness of the encounter, with *Omo* this strangeness remains; it is indeed the essential quality of the encounter.

Transparency and consistency are commonly lauded as principles of design that make products accessible by way of their predictability. The design of *Omo* seems to run counter to this principle: *Omo* is not regular in its behavior and interactions with others. The robot responds in one way at times and another way at other times. At times, it mimics the breathing patterns of its human companion, and at other times, it makes abrupt and startling changes in its own breathlike movements. As a social robot, *Omo* presents a kind of companionship and affect different from PARO or PaPeRo. Its companionship is not altogether subservient and is designed to include irrationality as a desirable feature, not as the flaw that Picard associates it with in her construction of affective computing. The remainder here is the irrationality of affective experience, and the design of *Omo* brings this often excluded quality of affect to the fore.

Agonistic Reification

Amy and Klara is a robotic system composed of two synthetic speech robots designed by Marc Böhlen (2006a).[20] According to Böhlen, the purpose of this system is to explore the expectation, construction, and maintenance of norms of public speech and the ways that those norms and our experiences with variations to those norms are affected when transferred to nonhuman entities that are mimicking humans (2006a, 2006b, 2008). This topic of exploration is not unusual in the context of human-robot interaction research. In that field, one can easily imagine social science experiments grounded in some notion of productive communication or cooperation, conducted under controlled circumstances with clear variables, resulting in empirical findings and design guidelines. Böhlen's work is differently situated, however. *Amy and Klara* is an experimental art and engineering project, and as such, the practices and agendas of conventional human-robot interaction research do not need to be upheld. With *Amy and Klara*, expectations of robot communication are explored not in the common form of social interaction but at one of the limits of human communication—cursing. By this exploration of the limits of the social qualities of robots and human-robot interaction, *Amy and Klara* is akin to *Blendie* and *Omo* in challenging our assumptions of human-robot relations, and it provides yet another example of an encounter with social robots that is made agonistic through the design of its embodiment.

The two synthetic speech robots *Amy and Klara* are physically instantiated as stationary boxes that are painted hot pink, equipped with speakers, and that curse at and argue with each other (figures 3.4 and 3.5). One of them (*Klara)* speaks with a German accent. Böhlen's choice of speech and cursing as the basis for these robots presents a critical perspective on computational speech recognition, speech generation, and adaptive dialog. As he explains (Böhlen 2006a),

> It is not only the disconnect between a human voice and a box that produces it that can make one feel uncomfortable. It is also what these voices have to say to us. The language of synthetic speech recognition and synthesis systems is a highly selective subset of the full, rich and messy body of linguistic corpora that comprise our oral and written languages. Exclamations are absent, questions are rare and the vocabulary is generally optimized for commerce.
>
> Here the charged world of foul language is under investigation. Swearing offers several interesting conduits into a critique of the under-exposed normative tendencies in automated language representation and social robotics. Why are most smart gadgets and toys friendly and playful, why are they usually modeled as pets or servants? Machines that curse and pick a fight might offer a more realistic preparation for a shared future between machines and humans.

Amy and Klara is agonistic in multiple ways. As is made clear in the preceding quote from Böhlen, the purpose of these robots is to question

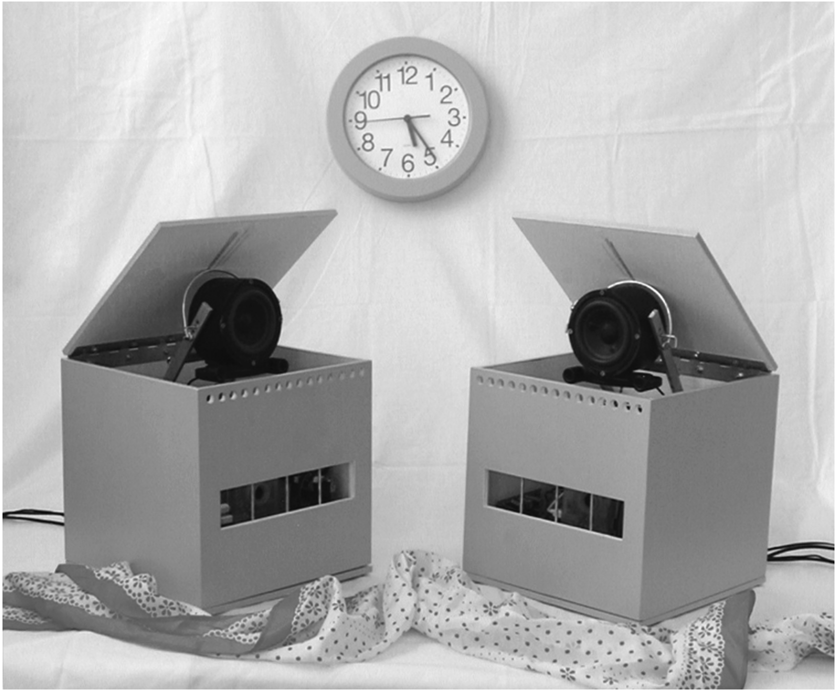

Figure 3.4
Marc Böhlen, *Amy and Klara* (2006a)

foundational assumptions of human-robot communication and language-based robot expression. As with *Omo* and *Blendie*, the social character of social robots is being explored—what kinds of communication are assumed proper and privileged and what kinds of communication are not and are thus left out of social robot design. The remainder in this case is cursing or what would commonly be considered abusive, juvenile, petty, or dysfunctional communication. Similar to *Omo*, the remainder with *Amy and Klara* is a mode of expression and interaction that falls outside of the rational and productive directives that tend to drive mainstream social robotics. More than simply documenting and representing issues of social robot design, *Amy and Klara* is a demonstrative, interactive instantiation of these issues.

The design of *Amy and Klara* leverages and explores computational text-to-speech, automated speech recognition, and aspects of computational vision as they figure into the construction of social robots. With *Amy and Klara* the substance of the robots' speech is produced by software that

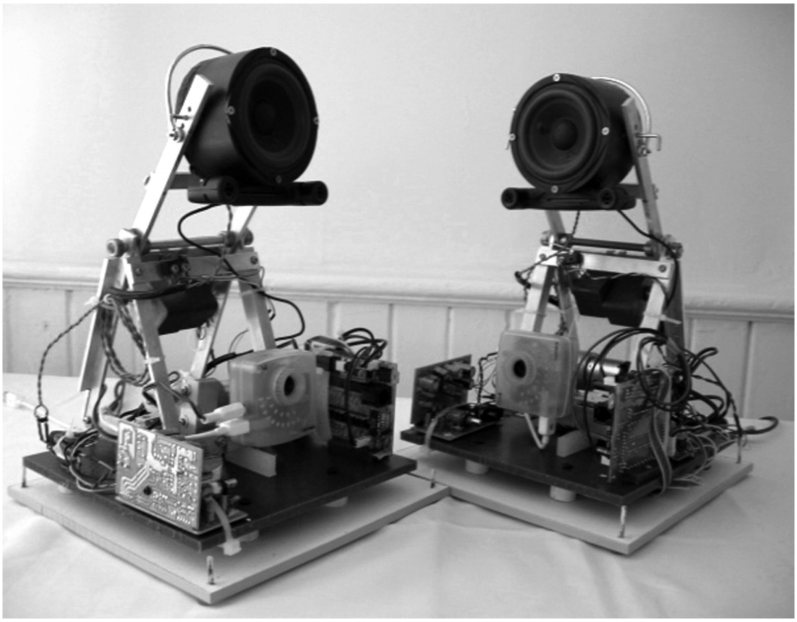

Figure 3.5
A view of the electronics of *Amy and Klara,* Marc Böhlen (2006a)

accesses and reads online lifestyle magazines. The content from these Web sites is parsed and becomes the basis for the construction of ontologies, or computational models of the world, possessed by the robots. As these linguistic models develop, certain words are given weight by virtue of their frequency. These words become the speech that the robots exchange. But sooner or later, there is a misunderstanding between the robots. When one robot fails to understand the other robot, it might be because it has developed a different ontology and thus is speaking a word the other robot does not know, or it might be because of the distortion inherent in the microphones and speakers, which is exacerbated by the German accent of *Klara*. Whatever the reason for the misunderstanding, "dissent arises and they begin to call each other names" (Böhlen 2008, 211). When one robot detects the foul language of the other, it responds in kind with an utterance containing foul language. These quarrelsome exchanges rapidly increase in intensity. As Böhlen (2008, 212) describes the basic procedural structure of the software regulating the social exchange:

Repeated use of a curse word from scale n leads to the selection of a curse word from scale $n + 1$, provided the following word is recognized as a curse word within a given time frame (otherwise the aggression levels recede). Since recognition and utterance occur in quick succession, both a low level exchange (when recognition results are poor) as well as a heated escalating fight, if recognition results are positive, are possible.

In addition to linguistic modes of exchange, the design of the robots' embodiment employs a computational vision system that regulates their interaction. Each robot is equipped with a camera that is pointed at the other robot, and the camera's field of view includes parts of the surrounding room. Once each robot has developed its corpus of words, and they have been catalogued and compared, the verbal exchange state is triggered by this vision system, which initiates the dialog when it identifies the color pink (that is, when it registers the presence of the other robot). The vision system of both robots is also able detect the presence of a human, as identified by overall shape. When a human is detected by the vision system, the robots change their behavior by first lowering their voices, curtailing their dialog with one another, and finally asking the people present to leave. So these social robots are designed to be social with each another but asocial with people.

The color pink plays an important role in the design of *Amy and Klara*. On the surface, the color pink serves as a gendering device. Together with the voice and the names, it works to construe the robots as prototypically feminine. But the color pink has another purpose beyond serving as a signifier for human understanding. In robotics, the color pink is often used as a test color for computational vision research. Because of the distinctive chromatic qualities of pink, specifically hot pink, it is commonly used for location and targeting purposes in computational vision. There are, for example, robot demos and competitions in which robots search for pink-colored tags in the environment or on people. The color pink thus does dual duty in the design of *Amy and Klara*. It genders the robot in human terms and is simultaneously a fundamental aspect of a robot-centric embodiment, enabling its vision in a manner that reflects the robot's distinctive sensing capacities.

Like *Blendie* and *Omo*, *Amy and Klara* evokes the uncanny. The primary way it does so is by the content and character of the dialog, which is at odds with the expressive capabilities of the robots. When people hear the exchanges between *Amy* and *Klara*, they do not mistake the dialog as occurring between humans. The prosidy of the speech is dull and dronelike. Due to the processing time involved for one robot to recognize the other's utterance, the timing between the exchanges feels fractured. The German accent of *Klara* amplifies the incommensurate quality of the scenario. Such an accent is associated with sternness, but in this context, the attempt to imbue the robot's voice with a greater amount of personhood only draws attention to the awkwardness of the technology in mimicking human dialog. The overall experience transforms the line between a human mode of communication and its replication into a fissure.[21]

In addition, Böhlen's framing of the dialog as fundamentally antagonistic is striking. Unlike other social robots, *Amy and Klara* are not designed for companionship or therapy; they are designed to engage in verbal conflict with one another. So even the relatively benign exchange that Böhlen (2006a) provides as an example in the project documentation is disconcertingly at odds with our preconceptions of what and how robots might verbally communicate:

R1: "Hey you."
R2: "Leave me alone." (synthetic German accent)
R1: "What is wrong with you?"
R2: "Leave me alone please." (synthetic German accent)

Finally, the design of *Amy and Klara* also challenges assumptions concerning social robots in another, more fundamental manner: they appear to be robots that exist primarily in relation to one another, with only a peripheral connection to humans. In this way, they confound expectations of social robots as intelligent artifacts that are social with humans. The social qualities of social robots serve primarily as a mode of robot-to-robot interaction, and the robots are social with one another, to the exclusion of humans. This disassociation from humans is made all the more strange by the fact that robots are nonhuman entities that depend on a human mode of communication with each other: they speak to each other. One result of this odd scenario is a further troubling of the common we/they distinctions between people and machines. In the case of *Amy and Klara,* the we and the they are conflated as the robots take on decidedly human qualities.

The design of *Amy and Klara* exemplifies a variation on the tactic of reconfiguring the remainder that I call *agonistic reification*. In what might first appear as a paradox, reification—the process of objectifying a thing—can be employed agonistically in the design of robots to produce encounters that objectify human beings in ways that prompt critical reflection on the processes and effects of objectification. The linguistic content coupled with the ways in which language is handled in the design of *Amy and Klara* provides an example. The use of a German accent to connote sternness plays on stereotypes, which are means of socially objectifying people. But this objectification extends social construction also to include the technical construction of the robot. The design of the robot's embodiment—the computational rendering of a synthetic voice into a German accent—requires objectifying the synthetic voice itself, making it into an element that can be examined and manipulated. In its original form, the synthetic voice does not have a German accent. The German accent is constructed from what is considered to be a more neutral American accent by procedurally "swapping select vowels and consonants" in the text-to-speech software. For example, "Welcome" is transformed to "¢velk2:m."[22] So Böhlen's exploration of the social qualities of speech (for example, expectations, norms, and stereotypes) is an exploration of the social *and* technical qualities of speech. Indeed, in the design of *Amy and Klara* and arguably all social robots, the exploration of the social depends on an exploration of the technical, in effect, melding these two into a single thing. Through such reciprocal plays on stereotypes, language content, and computational phoneme manipulation, the design of *Amy and Klara* fulfills both parts of the definition of objectify: it transforms a quality or condition into a unit of analysis, and it makes that quality or condition actual—that is, materially instantiated and experiential.

Reification in this case serves the agonistic purpose of highlighting assumptions concerning human qualities and relations and the way that those assumptions are imbued in computational systems. That is, reification is one way of bringing the remainder to the fore. Selecting and transforming a particular quality into unit of analysis provides a means to examine and manipulate that quality or unit.[23] With *Amy and Klara*, dialogic interaction is reified, examined, and manipulated. This occurs through a nesting or layering of objectification. Dialogic interaction becomes reified as cursing, and that particular irrational mode of human speech exchange is reified into a series of stimuli and response mechanisms and procedures, which are reified through the discrete computational manipulation of speech.

As with the prior examples of engineering the uncanny, this agonistic encounter is achieved through the design of the robot's embodiment. But contrary to the prior examples of *Blendie* and *Omo* in which embodiment was a quality of a robot that coupled it with humans, embodiment here is a quality of a robot that couples it to another robot. Even the pink coloring of the robot eludes simple explanation as a means for expressing human gender norms: it functions to enable coupling between the computational objects themselves. Thus *Amy and Klara* extend the machine-centric point of view witnessed in *Blendie* to the point of hyperbole. The design of the robot's embodiment operationalizes the human as a base form of stimulus. Such agonistic reification can be cast as an ironic but still critical response to concerns about translating human qualities into machines. To use Suchman's term, the tactic of reification performs a kind of "retrenching." This retrenching or performance of reification, however, is done in a manner that is self-mocking and contradictory, as it reduces sociability to a single, bounded feature and then instantiates that feature in robot form to produce an exaggerated effect (or affect, as the case might be). In this case, the irony of *Amy and Klara* demonstrates in material and experiential form the problems of extending machine sociability in human terms, and *Amy and Klara* functions as incitement for reflection—to consider assumptions in designing sociability through robot embodiment. For example, one assumes that one's social interactions with robots will be congenial. Moreover, one assumes that social robots will be social with people. *Amy and Klara* demonstrate that there is no essential basis to those assumptions. There is no technical or social reason why social robots must behave that way. In considering how we will comingle with intelligent systems and the role of design in shaping those experiences, *Amy and Klara* is a test case of a scenario in the extreme. The two robots counter saccharine modes of expression and interaction by privileging the abusive and petty, and in doing so they provide another vector along which to consider the possibilities and limitations of designing sociability into computational entities.

Summary

Social robots offer another opportunity for examining what it means to do design with computation. An analysis of the design of social robots through the frame of agonism provides more examples of how the distinctive qualities of computational objects might be manipulated to evoke and explore political issues. In this case, the quality is embodiment, and the political issue regards future human-robot relations: what will be the character of these relations, and what qualities of the social are privileged or veiled by design? As research and development initiatives forge ahead with visions of social robots for partnership, companionship, and therapy, it is important to pause to consider the base assumptions of those projects. The question of how to interact with robots or other intelligent artifacts and systems is anything but settled. The possibilities for sources, kinds, and effects of embodiment are anything but exhausted. As the examples in this chapter elaborate, an agonistic approach to the design of social robots looks to embodiment as a means for offering alternatives that keep the space of design and our expectations of social robots open and available to dispute.

Commonly, design is a means to demarcate and advance certain perspectives concerning what is desirable in our present and future. The therapy robot PARO is a case in point, as it materializes beliefs concerning the role of robots in society and the ways we might engage them. But just as design can set boundaries, it can also be used agonistically to disturb those boundaries. This disturbance occurs not by the erasure of lines of difference but by introducing productively disruptive tangents. The robots *Blendie*, *Omo*, and *Amy and Klara* are examples of such productively disruptive tangents. They highlight anxieties with technology, offer new modes of affective and intimate interaction, and bring to the fore assumptions concerning human expression and communication with animated artifacts.

On the surface, the political issues of social robot design appear quite different from the political issues described with regard to information design in chapter 2. Whereas the political issues related to information design were obvious and direct—issues of military funding and academic research, the price of oil, networks of corporate influence—with social robots, they are not. This difference is valuable because it shows the range of agonistic endeavors: adversarial design is not limited to what we commonly consider political issues and certainly not limited to ideological frames of left and right, conservative or liberal. In fact, revealing and articulating the contestable aspects of situations often perceived as nonpolitical is a central goal of agonism because the political is a pervasive condition and the contention that characterizes agonism should occur continuously and everywhere.

Addressing assumptions and expressing alternatives begins to engage what Mouffe (2000a, 2005b) refers to as contingency of the social order—that things could always be otherwise and every order is predicated on exclusion. Moreover, it calls attention to what Honig (1993) terms the remainder—the thing that is excluded. In this case, the “things that could be otherwise” are characteristics of relationships between people and robots. Through the design tactic of reconfiguring the remainder, what is excluded is brought to the fore and made materially and experientially available via the robot’s embodiment. The analysis of *Blendie*, *Omo*, and *Amy and Klara* shows that each can be interpreted as critical expressions of the remainders of social robot design, specifically anxiety and the irrational.

Furthermore, these agonistic encounters with robots are also political in that they envisage the possibility of different we/they relations—another core task to agonism (Mouffe 2005b). Whereas in political theory discussions of we/they distinctions are usually construed as distinctions between socioeconomic classes or categories of people, here it is initially construed as the relations between people and robots. But these agonistic encounters with social robots do not settle that we/they relation. The familiar distinctions between the categories of human and robot are not defended or upheld. Instead, the designs of these robots trouble, even confound, the assumptions on which these distinctions are commonly made. This troubling of categories is a base activity of adversarial design because it leads to exposing the remainder and thereby the political issues that inhabit and transect those categories. In the case of social robot design, the political issue is not whether but *how* humans and robots will interact. The agonistic endeavor is not to keep the categories of human and robot separate, but rather to identify and explore how the qualities of those categories might intermingle, and through design, to probe the possible relations between people and intelligent artifacts.